

This chapter covers the following topics related to Objective 1.3 (Summarize cloud concepts and connectivity options) of the CompTIA Network+ N10-009 certification exam:

- Network functions virtualization (NFV)
- Virtual private cloud (VPC)
- Network security groups
- Network security lists
- Cloud gateways
  - Internet gateway
  - Network address translation (NAT) gateway
- Cloud connectivity options
  - VPN
  - Direct Connect
- Deployment models
  - Public
  - Private
  - Hybrid
- Service models
  - Software as a service (SaaS)
  - Infrastructure as a service (IaaS)
  - Platform as a service (PaaS)
- Scalability
- Elasticity
- Multitenancy

CHAPTER 3

Cloud Concepts

It is not just a fad. It is not just marketing hype. Cloud computing is the future, and it is exploding in usage and usefulness. In this chapter, you will learn about some of the most foundational and critical topics in the vast field of cloud computing today.

Foundation Topics

Network Functions Virtualization (NFV)

One of the technologies that has helped cloud computing become a very popular reality is virtualization. While virtualization of servers was the initial push, these days we are seeing virtualization move into all aspects of Information Technology (IT). ***Network functions virtualization (NFV)*** is a great example of this. NFV abstracts and virtualizes important network functions that would traditionally be implemented by physical network appliances. Examples of network functions that can be virtualized with NFV are numerous and include

- Routing
- Switching
- Firewalling
- Load balancing
- Content delivery networking (CDN)
- Network monitoring

Thanks to NFV, the complex functionalities of the preceding network functions are decoupled from proprietary hardware and implemented as software applications. While this fact alone presents many advantages, IT administrators love the fact that they can run these applications on standard commercial off-the-shelf (COTS) servers. This decoupling of the network functions from specific hardware enables network operators to dynamically deploy, manage, and orchestrate network services with greater agility and efficiency, while also facilitating the optimization of resource utilization and operational costs. Specifically, just some of the advantages of NFV include

- **Doing more with less:** NFV typically permits dramatic increases in workload capacity with less power consumption, a smaller data center footprint, and reduced cooling requirements.
- **Reduced vendor lock-in:** Because NFV fosters the use of COTS servers, organizations can avoid being locked into proprietary hardware solutions from a single vendor.
- **Increased flexibility:** NFV tends to increase the agility of network operations because NFV facilitates quick changes to the network infrastructure, especially when compared to the traditional, nonvirtualized environments.

Cloud Networking Components

If an organization is interested in building IT solutions in the cloud, it is going to need key networking components and is also going to want to easily secure these resources. In this section, let's examine how an organization can accomplish both of these requirements with the cloud services of a leader in cloud technology—Amazon Web Services (AWS).

NOTE While this section focuses on AWS-specific examples, it is important to realize that other major cloud providers offer similar services. The specific names just vary.

Virtual Private Cloud (VPC)

The AWS ***virtual private cloud (VPC)*** lets you provision a logically isolated section of the AWS Cloud where you can launch AWS resources in a virtual network that you define. You have complete control over your virtual networking environment, including the selection of your IP address range, creation of subnets, and configuration of route tables and network gateways. You can use both IPv4 and IPv6 in VPCs for secure and easy access to resources and applications. Figure 3-1 shows elements inside a VPC.

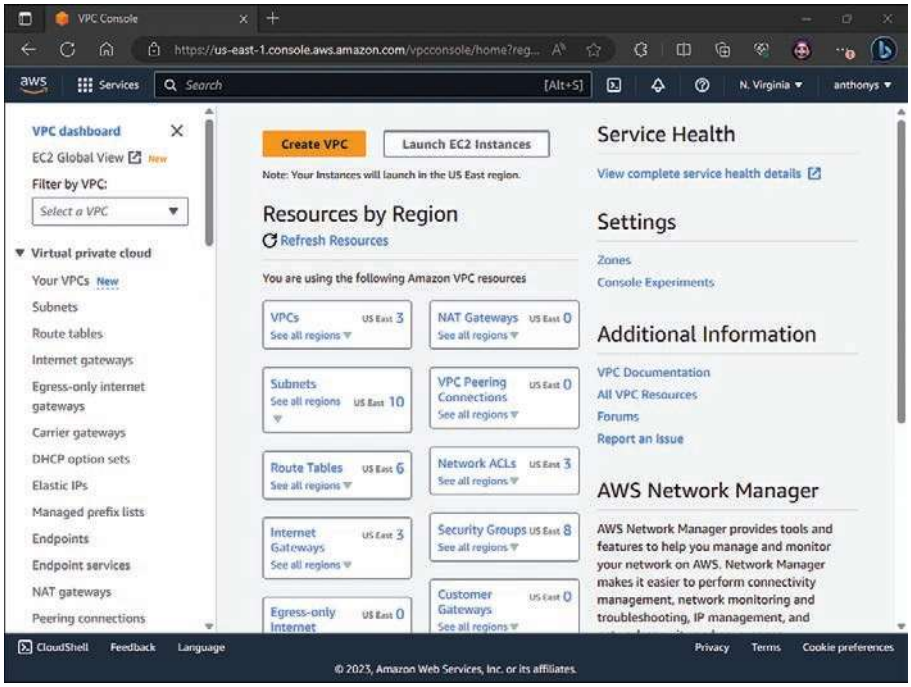


Figure 3-1 Some Components of an AWS VPC

NOTE AWS is not the only public cloud vendor to offer virtual private cloud functionality. Other major public cloud providers like Google Cloud Platform (GCP) and Microsoft Azure allow VPC functionality. Be aware that in Microsoft Azure, the VPC-like service is named Virtual Networks.

Network Security Groups

Because the security of resources in the cloud is a prime concern for both cloud customers and cloud providers, it is no big surprise that AWS provides built-in firewalls for use with its virtual compute resources. These ***network security groups*** help you easily control the accessibility of your virtual machines and their resources. Figure 3-2 shows an example of a network security group (named sg-f76c8d80) assigned to a virtual network interface in AWS.

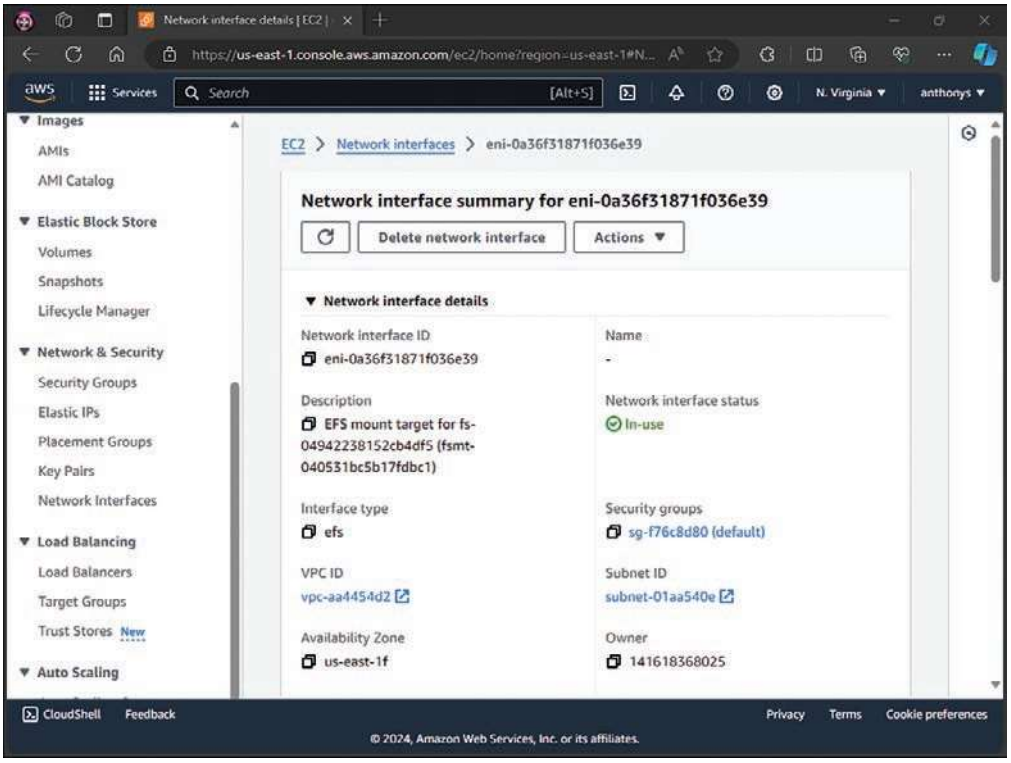


Figure 3-2 A Security Group Assigned to a vNIC

NOTE AWS refers to network security groups as simply security groups and often abbreviates them as SGs.

Consider the case of a web tier available in the AWS VPC. The AWS customer can configure the security group for this tier to permit HTTP and HTTPS traffic from customers using the web tier, and at the same time they can permit their team of support engineers to access the web tier using SSH and RDP. All other protocol attempts at accessing the web tier are denied by the security group.

Note that AWS applies a security group by attaching the security group to the virtual network interface (vNIC) of a virtual machine. This allows the security group to easily control (firewall) the traffic entering and exiting the virtual machine instance.

Network Security Lists

To further enhance cloud security, AWS also implements a ***network security list*** functionality (like other cloud vendors). AWS specifically terms its solution *network access control lists (NACLs)*. NACLs allow cloud customers to control access to their VPC subnets. NACLs are stateless constructs, which means customers must configure inbound and outbound rules, as there is no automatic recognition of state with traffic flows, and there are no automated access entries.

Students are often puzzled about why these "firewalls" exist when there are already security groups inside AWS. Remember that security groups are attached to virtual network interfaces of virtual machine instances and can control traffic to and from the virtual machines. NACLs are attached to subnets in a VPC to control traffic into and out of the entire subnet. Having multiple levels of built-in firewalls within the AWS network infrastructure permits fine-grained control of security.

Figure 3-3 shows an example of a NACL inside AWS.

Cloud Gateways

A ***cloud gateway*** is network infrastructure component or service that provides secure and efficient connectivity between on-premises networks or devices and cloud-based resources, applications, or services. Cloud gateways act as intermediaries or bridges, facilitating communication, data exchange, and access control between the local network and cloud environments, such as public cloud platforms, software as a service (SaaS) applications, or hybrid cloud deployments.

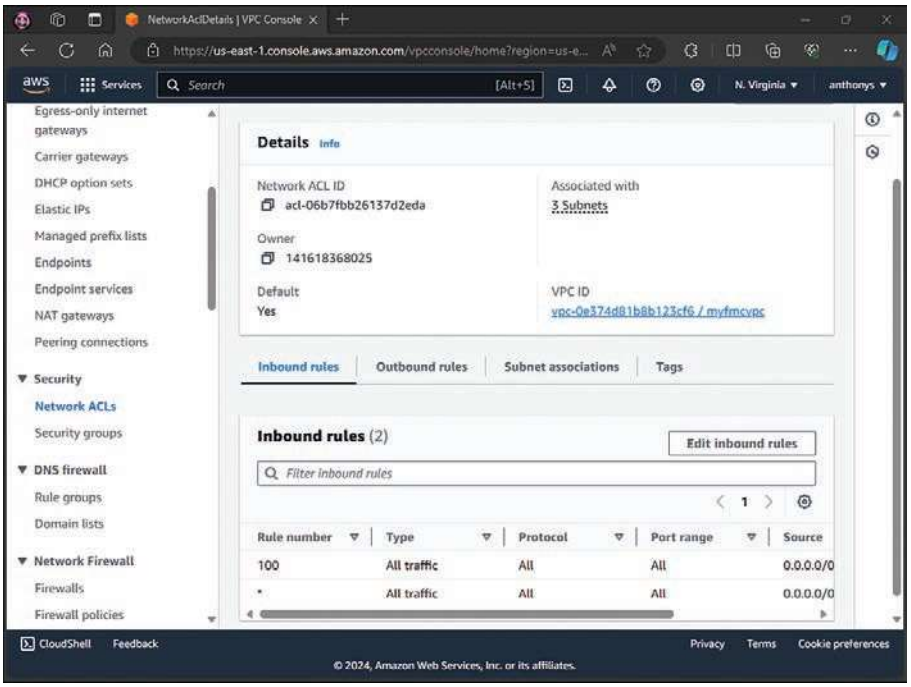


Figure 3-3 A NACL in AWS

There are several different types of specific cloud gateways commonly seen in cloud environments. These include the following:

- ***Internet gateway*:** An Internet gateway serves as a pivotal entry and exit point for network traffic between the cloud environment and the broader Internet. As a virtualized component within the AWS infrastructure, the internet gateway seamlessly connects resources hosted in virtual private clouds to external networks. This connectivity enables bidirectional communication while ensuring security and scalability. By facilitating the flow of data between AWS services and the Internet, the Internet gateway allows organizations to build, deploy, and manage cloud-based applications with flexibility and efficiency, while also providing robust access control mechanisms and integration options for establishing secure connections with on-premises infrastructure.
- ***Network address translation (NAT) gateway*:** While an Internet gateway in AWS seamlessly facilitates Internet connectivity, it does not provide fine-grained control over the network address translations required to translate private-use-only IP addresses with publicly routable IP addresses. Should a customer want to control these NAT translations, AWS (and its competitors) offers a NAT gateway. Acting as a managed service, the NAT gateway

translates outgoing traffic from private IP addresses of instances within the VPC into public IP addresses, allowing them to communicate with external networks while concealing their internal addresses.

Deployment Models

How do we implement (or deploy) cloud services most often today? In this section, we examine several different deployment models for cloud. Virtualized services and solutions are often offered by service providers as *cloud computing*. A company purchasing cloud computing services has several options to choose from:

Key Topic

- ***Private cloud***: With this model, the cloud infrastructure is provisioned for exclusive use by a single organization comprising multiple consumers, which might be business units. It might be owned, managed, and operated by the organization, a third party, or some combination of both, and it might exist on or off premises. The key is that this cloud is not for use by multiple organizations.
- ***Public cloud***: With this model, the cloud infrastructure is provisioned for open use by the general public. It might be owned, managed, and operated by a business, an academic institution, a government organization, or some combination of the three. It exists on the premises of the cloud provider, and these premises are typically located all over the globe to facilitate reduced latency from any location. This is the model presented by AWS and its main competitors, Microsoft Azure and Google Cloud Platform.
- ***Hybrid cloud***: With this model, the cloud infrastructure is a composition of two or more distinct cloud infrastructures (private or public) that remain unique entities but are bound together by standardized or proprietary technology that enables data and application portability. This is a widespread deployment model today. For example, an organization might build a private cloud for operational financial transactions while relying on a public cloud model for encrypted archiving of legacy transactions that must be maintained for compliance reasons.

NOTE While the main cloud deployment models are private, public, and hybrid, there is also the ***community cloud*** model. Community cloud services are used by individuals, companies, or entities with similar interests. A community cloud is typically a partnership between many different cloud vendors or organizations. An excellent example of a community cloud is cloud.gov in the United States. This community cloud provides a wide variety of services to employees of the U.S. government. These services are offered in the community cloud by several different public cloud vendors.

When an organization requires scalability, wants reduced costs, lacks in-house administrative personnel trained in cloud technologies, and has a complex network, a public cloud is the best deployment model to use. The advantages of using a public cloud include the following:

- Lower infrastructure, maintenance, and administrative costs
- Greater hardware efficiency
- Reduced implementation times
- Availability of short-term usage

When an organization needs to maintain strict control of business-critical data or is in a highly regulated businesses, such as the financial industry, a private cloud is typically the best choice.

A hybrid cloud environment is the best choice when an organization offers services that need to be configured for diverse vertical markets or has varying needs. Hybrid clouds are also ideal for organizations migrating to public clouds because they can ensure interoperability and efficiencies across the different environments.

Service Models

Cloud services are typically offered using one of several "as a service" models, which are usually offered on a pay-as-you-go basis. This is similar to the model used with utilities in the home (for example, electricity, Internet).

The following are a few of the service models that are available as part of cloud computing:

Key Topic

- ***Infrastructure as a service (IaaS)*:** With IaaS, the company rents virtualized servers (which are hosted by a service provider) and runs specific applications on those servers. IaaS is made simple thanks to public cloud providers like AWS, Microsoft Azure, and Google Cloud Platform.
- ***Software as a service (SaaS)*:** With SaaS, the details of the servers are hidden from the customer, and the customer's experience is similar to the experience of using a web-based application. Examples of SaaS in use today include Gmail, Microsoft 365, and Google Docs.
- ***Platform as a service (PaaS)*:** PaaS provides a development platform for companies that are developing applications and want to focus on creating the software without having to worry about the servers and infrastructure that are being used for that development. AWS Elastic Beanstalk and Google App Engine are examples of public cloud PaaS offerings.

NOTE Countless "as a service" designations have been established. For example, ***desktop as a service (DaaS)*** refers to cloud-hosted OS desktops made available to cloud clients. The term *everything as a service (XaaS)* is often used to describe the vast array of cloud services available now that the use of cloud approaches has exploded in popularity.

Key Cloud Concepts

We could easily fill a book this size with details of cloud technology, but for the CompTIA Network+ exam, you need to know the following key aspects:

Key Topic

- Cloud connectivity options
- Multitenancy
- Elasticity
- Scalability

Cloud Connectivity Options

As more and more technology has appeared in the cloud (both public and private), secured connectivity has become crucial. Secure protocols such as HTTPS, TLS, and SSH are necessary when accessing most cloud resources. Fortunately, massive cloud providers such as AWS make it simple to securely connect using a wide variety of methods, including hardware ***virtual private network (VPN)*** appliances located at your corporate or home office. Although your applications and servers might be sharing physical equipment with other customers of AWS, great pains are taken to ensure that this multitenancy (described in the next section) does not come at the cost of security.

AWS and other large public cloud providers go a step further, allowing customers to purchase AWS ***Direct Connect*** circuits. These circuits are private, dedicated leased lines directly into the AWS infrastructure. When you use AWS Direct Connect, your traffic does not commingle with that of other AWS customers or other Internet users at all. Of course, these types of connections are more expensive than other options.

NOTE The first thought most network engineers have when thinking about a service like AWS Direct Connect is the added security. Another huge benefit, however, is the predictable, low-latency performance such a private connection can offer.

Multitenancy

If you ask engineers why they are hesitant to move to the public cloud, they are likely to say that they are worried about security. When they react this way, it is almost always because they are thinking about ***multitenancy***. This is a fancy term for the fact that the physical servers in the public cloud infrastructure are hosting workloads for many different customers. Your virtual machines (VMs) might be located on the same physical server as the VMs of your biggest competitor. Multitenancy requires you to have great faith that the cloud provider has done excellent setup of security work to ensure that the VMs of one customer are completely hidden from other customers.

NOTE When you have the money to spend and need the certainty, you can purchase a fully dedicated hosting solution. In such a situation, the cloud provider leases to you your own dedicated hardware in its cloud—and this hardware is not shared with other customers.



Elasticity

Elasticity is another extremely compelling reason to move to the cloud. Elasticity means that a cloud solution can scale (up or down or in or out) as needed based on demand. Scaling up or down means adding or subtracting resources to provide larger or smaller systems. Scaling in or out refers to allowing systems to clone themselves or terminate themselves, as needed.

NOTE The National Institute of Standards and Technology (NIST) includes rapid elasticity as one of the five key characteristics that define what *cloud* truly means. The other four characteristics are on-demand self-service, broad network access, resource pooling, and measured service. If you are interested in reading this important document for yourself, check out NIST SP 800-145.

Elasticity is truly incredible. When your cloud offers you rapid elasticity (and it should!), your network infrastructure and services offered can grow and shrink along with demand. This is incredibly exciting for many reasons, but perhaps the two biggest reasons are the fact that you do not have to worry (as much) about your infrastructure failing under a large load, and you do not have to worry about overprovisioned resources when demand is low.

Scalability

It should be no surprise that if your cloud solution can grow and shrink as needed (elasticity), it has no problem demonstrating great scalability. ***Scalability*** refers to a solution's ability to grow with need or demand. Cloud technology today can do even better than this: It can grow or shrink as needed.

Real-World Case Study

Acme, Inc. is considering a move to the cloud with some key services it is having problems scaling from its on-prem, local data center. Specifically, Acme, Inc. is considering a public cloud implementation for the streaming of new training videos globally. Acme, Inc. needs these movies to be delivered in a format that is based on the bandwidth of the access device.

To facilitate this new public cloud initiative, IT personnel from Acme, Inc. have created a virtual private cloud in AWS and constructed the necessary infrastructure to support the deployment and conversion of video content globally. Acme is thrilled to be taking advantage of a hybrid cloud deployment model. It is utilizing its on-premises resources right alongside its cloud resources to facilitate a much-improved global delivery network.

Acme, Inc. is also investigating the changes in the security posture of the organization with this move. Specifically, Acme is exploring its security responsibilities in this new solution compared to the security responsibilities of the cloud provider. Acme personnel are carefully constructing security groups and network access control lists to accommodate careful, secured access of cloud resources.

Summary

Here are the main topics covered in this chapter:

- This chapter covered network functions virtualization and described this emerging trend in detail.
- This chapter described some of the key cloud networking components found today, including virtual private clouds, network security groups, network security lists, and cloud gateways.
- This chapter covered the deployment models often seen with cloud in networking today. These include private, public, hybrid, and community clouds.
- This chapter also covered the service models seen today with cloud, including SaaS, IaaS, and PaaS.
- Finally, this chapter covered key cloud concepts you should be aware of, including connectivity options, multitenancy, elasticity, and scalability.

Exam Preparation Tasks

Review All the Key Topics

Review the most important topics from this chapter, noted with the Key Topic icon in the outer margin of the page. Table 3-1 lists these key topics and the page number where each is found.

Key Topic

Table 3-1 Key Topics for Chapter 3

Key Topic Element	Description	Page Number
List	Deployment models	71
List	"As a service" models	72
List	Key cloud concepts	73
Section	Elasticity	74

Define Key Terms

Define the following key terms from this chapter and check your answers in the Glossary:

cloud gateway, community cloud, desktop as a service (DaaS), Direct Connect, elasticity, hybrid cloud, infrastructure as a service (IaaS), Internet gateway, multitenancy, network address translation (NAT) gateway, network functions virtualization (NFV), network security group, network security list, platform as a service (PaaS), private cloud, public cloud, scalability, software as a service (SaaS), virtual private cloud (VPC), virtual private network (VPN)

Additional Resources

Google Cloud Platform: Deploying Cloud Marketplace Solutions to Google Kubernetes Engine (GKE): https://www.ajsnetworking.com/gcp-compute/

Cloud Versus Outsourcing: https://www.ajsnetworking.com/comptia-cloud-essentials-cloud-vs-outsourcing/

Review Questions

The answers to these review questions appear in Appendix A, "Answers to Review Questions."

1. What cloud component is assigned to a virtual machine and acts as a firewall, controlling traffic into and out of the VM?
   a. Virtual private cloud
   b. Network security list
   c. Network security group
   d. Internet gateway

2. What type of cloud deployment model features many customers consuming resources from a single provider?
   a. Public
   b. Private
   c. Community
   d. Hybrid

3. What type of "as a service" model features a powerful cloud environment targeted at developers who need to test and deploy updates to their applications?
   a. IaaS
   b. SaaS
   c. PaaS
   d. DaaS

4. Which cloud characteristic refers to the ability to dynamically scale resources as needed during times of great demand as well as in times of low demand?
   a. Scalability
   b. Elasticity
   c. On-demand
   d. Centralized

5. Which of the following are popular options when it comes to ensuring security of cloud-based data in transit? (Choose two.)
   a. Public Internet
   b. VPN

c. Direct private connection
d. RDP

6. Which term refers to your company's data being stored on shared physical servers in the public cloud infrastructure?
   a. Hybrid
   b. Multitenancy
   c. Elasticity
   d. Orchestration

7. Which of the following enables you to provision a logically isolated section of the AWS Cloud where you can launch AWS resources in a virtual network?
   a. VPC
   b. NFV
   c. Internet gateway
   d. NAT gateway

8. Which of the following are the best examples of SaaS?
   a. AWS, Microsoft Azure, and Google Cloud Platform
   b. AWS Elastic Beanstalk, Google App Engine, and IBM Cloud
   c. AWS S3 and Azure Virtual Networks
   d. Gmail and Microsoft 365

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This chapter covers the following topics related to Objective 1.4 (Explain common networking ports, protocols, services, and traffic types) of the CompTIA Network+ N10-009 certification exam:

- File Transfer Protocol (FTP): Ports 20/21
- Secure File Transfer Protocol (SFTP): Port 22
- Secure Shell (SSH): Port 22
- Telnet: Port 23
- Simple Mail Transfer Protocol (SMTP): Port 25
- Domain Name System (DNS): Port 53
- Dynamic Host Configuration Protocol (DHCP): Ports 67/68
- Trivial File Transfer Protocol (TFTP): Port 69
- Hypertext Transfer Protocol (HTTP): Port 80
- Network Time Protocol (NTP): Port 123
- Simple Network Management Protocol (SNMP): Ports 161/162
- Lightweight Directory Access Protocol (LDAP): Port 389
- Hypertext Transfer Protocol Secure (HTTPS): Port 443
- Server Message Block (SMB): Port 445
- Syslog: Port 514
- Simple Mail Transfer Protocol Secure (SMTPS): Port 587
- Lightweight Directory Access Protocol over SSL (LDAPS): Port 636
- Structured Query Language (SQL) Server: Port 1433
- Remote Desktop Protocol (RDP): Port 3389
- Session Initiation Protocol (SIP): Ports 5060/5061
- Internet Protocol (IP) types
  - Internet Control Message Protocol (ICMP)
  - Transmission Control Protocol (TCP)
  - User Datagram Protocol (UDP)
  - Generic Routing Encapsulation (GRE)
  - Internet Protocol Security (IPsec)
    - Authentication Header (AH)
    - Encapsulating Security Payload (ESP)
    - Internet Key Exchange (IKE)
  - Traffic types
    - Unicast
    - Multicast
    - Anycast
    - Broadcast

CHAPTER 4

Networking Ports, Protocols, Services, and Traffic Types

This chapter might initially look like a reference chapter, and it can certainly serve that function, but this chapter is much more than that. This chapter also helps prepare you for more in-depth explanations of many of these protocols in later chapters of this book. Additionally, this chapter teaches you about the various IP protocol types in use today and the traffic types that help make our networks so incredibly useful.

Foundation Topics

Ports and Protocols

I know it might be very intimidating to look at this section of the chapter and realize just how many different protocols (and ports) are needed in a typical network today. Resist the urge to panic! Through study and practice, you will master these different protocols and the ports they use.

As you will quickly learn in this chapter, there are several common networking ports that are used frequently. Ports 0 through 1023 are defined as well-known ports. Registered ports are from 1024 to 49151. The remainder of the ports from 49152 to 65535 are used dynamically by applications.

FTP (File Transfer Protocol)

File Transfer Protocol (FTP) can be used to transfer files with a remote host (which typically requires authentication of user credentials). It uses TCP ports 20 and 21 in its operation. Unfortunately, FTP users authenticate using cleartext mechanisms. It is one of many protocols that lacks security (and is becoming rarely used as a result).

SFTP

Secure File Transfer Protocol (SFTP) provides FTP file transfer service over an SSH connection. Just like SSH, this secure protocol uses TCP port 22 in its operation.

SSH

Secure Shell (SSH) is used to securely connect to a remote host (typically via a terminal emulator). SSH has become the new de facto standard method of remote access for devices that use a command-line interface. SSH uses TCP port 22 in its operation.

Telnet

Telnet is used to connect to a remote host (typically via a terminal emulator). This remote access protocol should be used only in a lab or practice environment because it does not provide security. The alternative to this protocol that you should use in production is SSH. Telnet uses TCP port 23 in its operation.

SMTP

Simple Mail Transfer Protocol (SMTP) is used for sending email throughout a network. This protocol uses TCP port 25 in its operation.

DNS (Domain Name System)

Domain Name System (DNS) is used to translate domain names, such as www.pearsonitcertification.com, into IP addresses, such as 165.193.123.44. DNS uses a hierarchical namespace that enables the database of hostname-to-IP address mappings to be distributed across multiple servers. DNS uses both TCP and UDP in its operation. The port used in both cases is port 53.

DHCP (Dynamic Host Configuration Protocol)

You can use ***Dynamic Host Configuration Protocol (DHCP)*** to dynamically assign IP address information (for example, IP address, subnet mask, DNS server IP address, and default gateway IP address) to a network device. DHCP uses UDP ports 67 and 68 in its operation.

TFTP

Trivial File Transfer Protocol (TFTP) transfers files with a remote host (and does not require authentication of user credentials). TFTP uses UDP port 69 in its operation.

HTTP

Hypertext Transfer Protocol (HTTP) retrieves content from a web server. This protocol operates using TCP port 80. Many web server administrators liked to move the protocol to port 8080 in an attempt to circumvent the massive number of attacks against the default port 80. HTTP lacks security, and as a result, today, it is hard to find websites that are running HTTP. HTTPS is now the de facto standard for web servers.

NTP

Network devices use ***Network Time Protocol (NTP)*** to synchronize their clocks with a time server (that is, an NTP server). This protocol relies on UDP port 123 in its operation.

SNMP

Simple Network Management Protocol (SNMP) is used to monitor and manage network devices. SNMP uses UDP ports 161 and 162 in its operation. You will learn many details about SNMP in Chapter 14, "Network Monitoring."

LDAP

Lightweight Directory Access Protocol (LDAP) provides directory services (for example, a user directory that includes username, password, email, and phone number information) to network clients. LDAP is used to access and query compliant directory services systems, such as Microsoft Active Directory. TCP port 389 is used in its operation.

HTTPS

Hypertext Transfer Protocol Secure (HTTPS) is used to securely retrieve content from a web server. This secure version of HTTP uses TCP port 443 in its operation. HTTPS was originally made possible by the ***Secure Sockets Layer (SSL)*** protocol. ***Transport Layer Security (TLS)*** is the latest version of this technology. Because TLS is closely related to SSL, we often simply say that HTTPS is made possible by SSL/TLS.

SMB

Server Message Block (SMB) is used to share files, printers, and other network resources. This protocol was originally used by Microsoft clients only. Since then, UNIX and Linux clients can install software to also use SMB for file sharing. Older versions of SMB used both UDP and TCP port 3020. Today, newer versions of SMB (after Windows 2000) use port 445 on top of a TCP stack in order to communicate over the Internet.

Syslog

Syslog is the standard used by network devices (and other computer systems) to report and send log messages on status information and health events to network administrators. In fact, to centrally store these messages, they are often sent to a Syslog server. This data is often called *machine data*. Storing and analyzing this information, which may include driver failures, device conflicts, read/write errors, timeouts, and bad block errors, can be very important. Syslog uses UDP port 514 in its operation.

SMTPS

Simple Mail Transfer Protocol Secure (SMTPS) uses TCP port 587 in its operation and is the secure version of SMTP. SMTPS uses Transport Layer Security to provide authentication of the communication partners along with data integrity and confidentiality by wrapping SMTP data in TLS. This is like how HTTPS wraps HTTP data inside TLS.

LDAPS

Lightweight Directory Access Protocol over SSL (LDAPS) is the secure version of LDAP. It operates over TCP port 636.

Structured Query Language (SQL) Server

SQL (Structured Query Language) is a standardized programming language used to manage and manipulate relational databases by performing tasks such as querying, updating, and managing data. You can use this SQL language via a specialized server to run powerful queries against data that is stored in databases. A ***Structured Query Language (SQL) Server*** can use TCP and UDP in its operation. Port 1433 was reserved for this Microsoft invention.

RDP

Remote Desktop Protocol (RDP) is a Microsoft protocol that allows a user to view and control the desktop of a remote computer. RDP is a presentation layer protocol that supports a Windows-based Remote Desktop Connection (RDC) between an RDP client (formerly known as Windows Terminal Client) and a server. This remote access technology uses both TCP and UDP port 3389 in its operation.

SIP

Session Initiation Protocol (SIP) is used to create and end sessions for one or more media connections, including Voice over IP (VoIP) calls. SIP can use both TCP and UDP ports 5060 and 5061 in its operation.

Protocol/Port Summary

Table 4-1 provides a summary of protocols and their assigned ports

Key Topic

Table 4-1 Port Assignments for Commonly Used Protocols

Protocol	Description	Port Assignment
DHCP (Dynamic Host Configuration Protocol)	Dynamically assigns IP address information (for example, IP address, subnet mask, DNS server IP address, and default gateway IP address) to a network device	UDP 67, 68
DNS (Domain Name System)	Resolves domain names to corresponding IP addresses	TCP/UDP 53
FTP (File Transfer Protocol)	An insecure protocol used to transfer files with a remote host (typically requires authentication of user credentials)	TCP 20, 21
HTTP (Hypertext Transfer Protocol)	An insecure protocol used to retrieve content from a web server	TCP 80
HTTPS (Hypertext Transfer Protocol Secure)	Used to securely retrieve content from a web server	TCP 443
LDAP (Lightweight Directory Access Protocol)	Provides directory services (for example, a user directory that includes username, password, email, and phone number information) to network clients	TCP 389
LDAPS (Lightweight Directory Access Protocol over SSL)	The secure version of LDAP	TCP 636
NTP (Network Time Protocol)	Used by a network device to synchronize its clock with a time server (NTP server)	UDP 123
RDP (Remote Desktop Protocol)	A Microsoft protocol that allows a user to view and control the desktop of a remote computer	TCP/UDP 3389
SFTP (Secure File Transfer Protocol)	Provides FTP file transfer service over an SSH connection	TCP 22
SIP (Session Initiation Protocol)	Used to create and end sessions for one or more media connections, including Voice over IP (VoIP) calls	TCP/UDP 5060, 5061
SMB (Server Message Block)	Used to share files, printers, and other network resources	TCP 445
SMTP (Simple Mail Transfer Protocol)	Used for sending email throughout the network	TCP 25
SMTPS (Simple Mail Transfer Protocol Secure)	Allows for secure communication between email clients and mail servers, ensuring that email messages are encrypted during transmission	TCP 587

Protocol	Description	Port Assignment
SNMP (Simple Network Management Protocol)	Used to monitor and manage network devices; SNMPv3 is the latest and most secure version of SNMP and allows for the use of strong cryptographic algorithms, such as HMAC-SHA, HMAC-MD5, and AES, to ensure data integrity and confidentiality	UDP 161, 162
SQL (Structured Query Language) Server	A server that specializes in SQL and is used to run powerful queries against data that is stored in databases	TCP/UDP 1433
SSH (Secure Shell)	Used to securely connect to a remote host (typically via a terminal emulator)	TCP 22
Syslog	The standard used by network devices (and other computer systems) to report on status information and events	UDP 514
Telnet	An insecure protocol used to connect to a remote host (typically via a terminal emulator)	TCP 23
TFTP (Trivial File Transfer Protocol)	Used to transfer files with a remote host (does not require authentication of user credentials)	UDP 69

Internet Protocol (IP) Types

As you have probably realized, there are many, many protocols that are considered part of the TCP/IP protocol suite. This section provides an overview of several of the IP protocol types used within TCP/IP. Many of these protocols are discussed in detail in other chapters.

Remember, ***Internet Protocol (IP)*** is a network layer protocol, documented in RFC 791, that offers a connectionless internetwork service. IP provides features for addressing, packet fragmentation and reassembly, type-of-service specification, and security.

Internet Control Message Protocol (ICMP)

In addition to TCP and UDP, ***Internet Control Message Protocol (ICMP)*** is another transport layer protocol you are likely to meet. ICMP is a network layer Internet protocol documented in RFC 792 that reports errors and provides other information relevant to IP packet processing. It is used by utilities such as ping and traceroute.

Transmission Control Protocol (TCP)

The transport layer of the OSI model offers an important protocol for reliable transport—***Transmission Control Protocol (TCP)***. What does *reliable transport* mean, in networking terms? It means that if a segment is dropped, the sender can detect that drop and retransmit the dropped segment. Specifically, a receiver acknowledges segments that it receives. Based on those acknowledgments, a sender can decide which segments were successfully received and which segments need to be transmitted again.

User Datagram Protocol (UDP)

User Datagram Protocol (UDP) can be considered an opposite approach to TCP. Unlike TCP, UDP offers unreliable transport. This means that if a segment is dropped, the sender is unaware of the drop, and no retransmission occurs. Note that UDP is another option at the transport layer of the OSI model. This works as an ideal alternative to TCP.

NOTE The reliability of TCP versus the unreliability of UDP is often referred to as ***connectionless*** versus ***connection-oriented*** communications. You might use the analogy of a phone call versus a postcard. With a phone call (which represents TCP), as you send messages, you ensure that the connection remains strong and that communications are clear. For example, you might ask "can you hear me now?" as you move to a different part of the house for a better cell connection. In fact, you begin the connection with a "hello" and end the connection with a "goodbye." UDP operates more like a postcard: There is no connection maintained at all. You drop the postcard in the mail and hope that it reaches its destination.

Generic Routing Encapsulation (GRE)

Generic Routing Encapsulation (GRE), as its name implies, is a multipurpose, simple tunneling protocol that you might call on in a wide variety of circumstances. For example, let's say you have traffic that you want to protect over an IPsec tunnel, but this traffic is multicast traffic. IPsec does not support securing this type of traffic. That is where GRE can come into play. The multicast can be encapsulated inside GRE, and then this GRE traffic can be protected by IPsec. This is just one example of many where GRE can come in handy.

Internet Protocol Security (IPsec)

Internet Protocol Security (IPsec) is a complex suite of protocols that is used to create secured connections between network systems. In fact, IPsec is quickly becoming the de facto standard for VPN connections. IPsec features the use of ***Authentication Header (AH)*** and ***Encapsulating Security Payload (ESP)***. As you would guess, AH handles authentication functions, while ESP takes care of encryption. IPsec also takes advantage of the ***Internet Key Exchange (IKE)*** protocol. IKE negotiates cryptographic keys and security parameters to facilitate the secure IPsec data transmissions over the Internet.

Internet Protocol (IP) Types Summary

Table 4-2 provides a summary of the TCP/IP suite protocols described so far in this chapter.

Key Topic

Table 4-2 Internet Protocol (IP) Types Summary

Protocol	Complete Protocol Name	Description
IPsec	Internet Protocol Security	A complex suite of protocols that is used to create secured connections between network systems.
GRE	Generic Routing Encapsulation	A tunneling protocol used to encapsulate a wide variety of network layer protocols inside virtual point-to-point links or point-to-multipoint links over an IP network.
IP	Internet Protocol	A network layer protocol, documented in RFC 791, that offers a connectionless internetwork service. IP provides features for addressing, packet fragmentation and reassembly, type-of-service specification, and security.
TCP	Transmission Control Protocol	A connection-oriented protocol that offers flow control, sequencing, and retransmission of dropped packets.
UDP	User Datagram Protocol	A connectionless alternative to TCP used for applications that do not require the functions offered by TCP.
ICMP	Internet Control Message Protocol	Sends error messages and operational information indicating issues with packet delivery in IP networks.

Protocol	Complete Protocol Name	Description
ESP	Encapsulating Security Payload	Provides for encryption in the IPsec suite of protocols.
AH	Authentication Header	Provides for authentication in the IPsec suite of protocols.
IKE	Internet Key Exchange	Negotiates cryptographic keys and security parameters for IPsec.

Traffic Types

In our IP-based networks of today, traffic flows in different manners as dictated by the various traffic types we create. In this section, we examine these different traffic types and how data moves throughout the network with each.

Unicast

Most network traffic is ***unicast*** in nature, meaning that traffic travels from a single source device to a single destination device. Figure 4-1 illustrates an example of a unicast transmission.

Key Topic

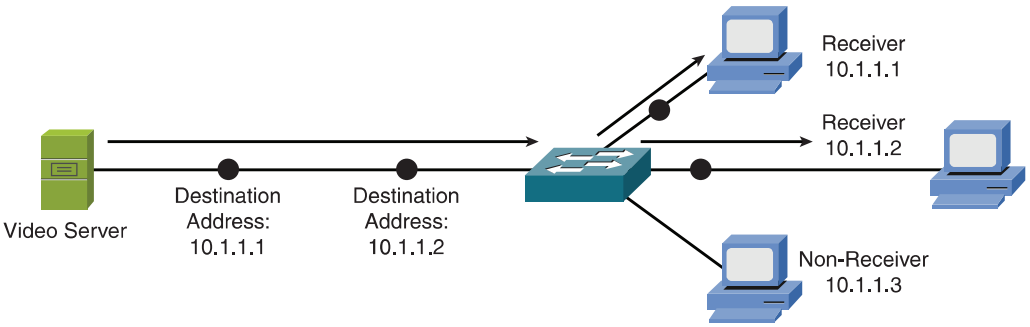


Figure 4-1 Sample Unicast Transmission

Notice the Layer 3 IP addresses shown in Figure 4-1. We delve deep into these addresses in Chapter 7, "IPv4 Addressing," but for now it is worth noting that these are unicast IP addresses to facilitate this most common form of IP traffic.

It is at this point that you might ask, "Why do we even need other forms of traffic in the IP network?" This is a great question. The answer lies with how inefficient unicast traffic would be when we have something (let's say a large video file) that we want to send to all devices on the network. Creating a separate copy of the video for every single recipient and then sending the data to every single network node would

be horribly inefficient. This is a big part of the reason why we have other network traffic types that we see in this section of the chapter.

Broadcast

Broadcast traffic travels from a single source to all destinations on a network (that is, a *broadcast domain*) as shown in Figure 4-2. Notice the use of a very special IP address to facilitate the broadcast traffic type—it is 255.255.255.255.

Key Topic

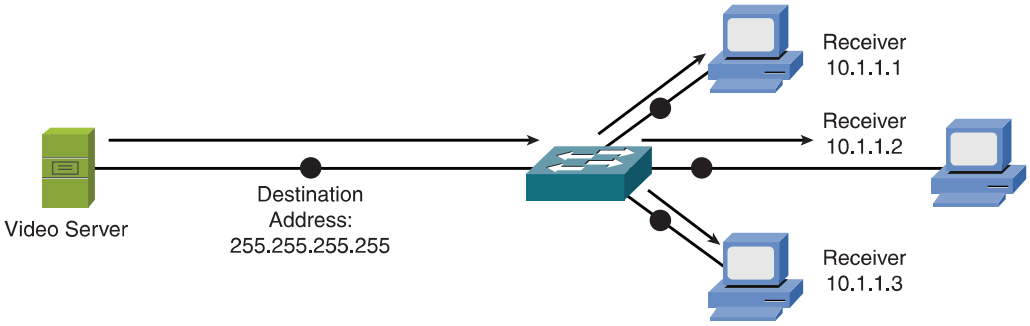


Figure 4-2 Sample Broadcast Transmission

It might seem as though the broadcast address 255.255.255.255 would reach all hosts on all interconnected networks. However, 255.255.255.255 targets all devices on a single network—specifically, the network local to the device sending a packet destined for 255.255.255.255. Another type of broadcast address is a *directed broadcast address*, which targets all devices in a remote network. For example, the address 172.16.255.255 /16 is a directed broadcast address targeting all devices in the 172.16.0.0 /16 network.

Multicast

Multicast technology offers an efficient mechanism for a single host to send traffic to multiple specific destinations. For example, say that a network has 100 users, and 20 of those users want to receive a video stream from a video server. With a unicast solution, the video server would have to send 20 individual streams, 1 for each recipient. As described earlier in this section, such a solution could consume a significant amount of network bandwidth and put a heavy processor burden on the video server.

With a broadcast solution, the video server would only have to send the video stream once; however, it would be received by every device on the local subnet, even devices not wanting to receive the video stream. Even though a lot of the devices do

not want to receive the video stream, they still must pause what they are doing and take time to check each of these unwanted packets.

As shown in Figure 4-3, multicast offers a compromise, allowing the video server to send the video stream only once and sending the video stream only to devices on the network that want to receive the stream. Multicast is possible thanks to the use of what is called a Class D IP address. A Class D address, such as 239.1.2.3, represents the address of a *multicast group*. The video server could, in this example, send a single copy of each video stream packet destined for 239.1.2.3. Devices wanting to receive the video stream could join the multicast group. Based on the device request, switches and routers in the topology could then dynamically determine out of which ports the video stream should be forwarded.

Key Topic

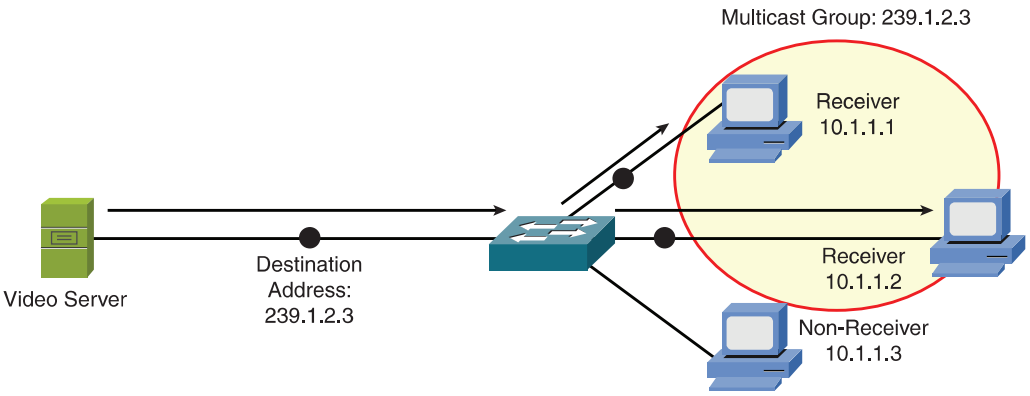


Figure 4-3 Sample Multicast Transmission

Are there other classes of IP addresses? There sure are, and you will learn all about them in Chapter 7.

Anycast

With ***anycast***, a single IP address is assigned to multiple devices, as illustrated in Figure 4-4. It is a one-to-nearest (from the perspective of a router's routing table) communication flow.

Notice the IP addresses shown in Figure 4-4, which might look quite odd to you. These are IPv6 addresses. As you will learn in Chapter 8, "Evolving Use Cases," IPv6 is the next-generation version of IP and it is becoming more and more common in our networks today. While anycast traffic is possible in IPv4, it is going to be much more commonly used in the new IPv6 protocol suite.

Key Topic

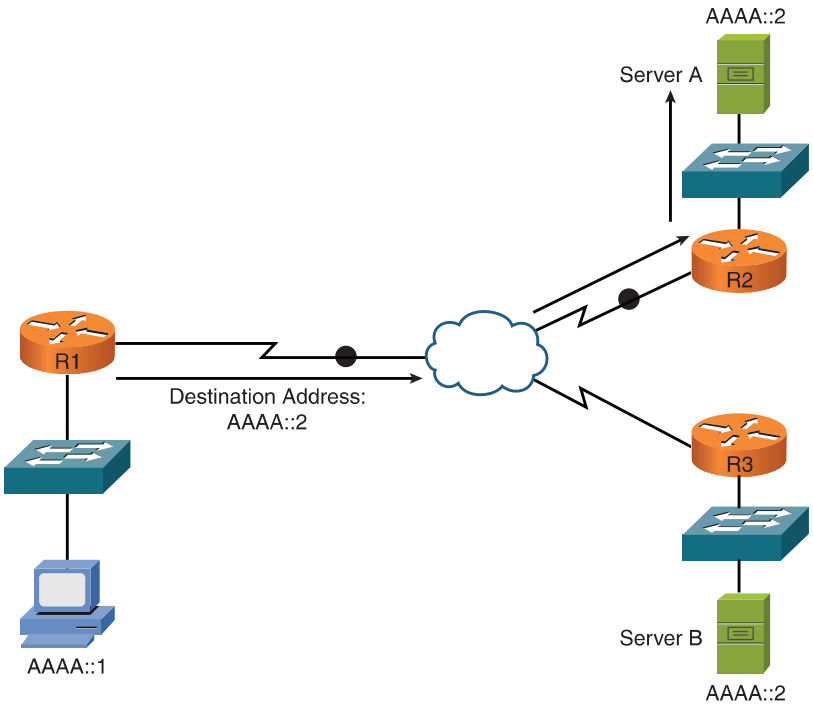


Figure 4-4 An IPv6 Anycast Example

In Figure 4-4, a client with IPv6 address AAAA::1 wants to send traffic to destination IPv6 address AAAA::2. Notice that two servers (Server A and Server B) have the IPv6 address AAAA::2. In the figure, the traffic destined for AAAA::2 is sent to Server A, via router R2, because the network on which Server A resides appears to be closer than the network on which Server B resides, from the perspective of router R1's IPv6 routing table.

Remember that you need to distinguish among these four network traffic types:

- With unicast, traffic is one-to-one communication between a single sender and a single receiver, and this is the most common type of traffic in IP networks. It is used for point-to-point communication between devices, such as client-server interactions and most Internet browsing.
- Multicast is a mechanism by which groups of network devices can send and receive data between the members of the group at one time (one-to-many), instead of separately sending messages to each device in the group.

- Anycast traffic is one-to-nearest communication from a sender to the nearest of multiple receivers. Multiple devices share the same anycast IP address, but packets are routed to the nearest (or best) destination based on routing metrics such as shortest path or lowest latency.
- A broadcast is at the opposite end of the spectrum from a unicast. Broadcast traffic is one-to-all communication from a single sender to all devices within a specific network segment. Each packet in a broadcast transmission is addressed to a special broadcast IP address (e.g., 255.255.255.255 in IPv4) or a Layer 2 broadcast address (e.g., MAC address FF:FF:FF:FF:FF:FF).

NOTE IPv6 was engineered to provide many advantages over the IPv4 protocol. One of the massive advantages is that broadcast frames and packets from IPv4 do not exist in an IPv6-only network. IPv6 uses only unicasts, multicasts, and anycasts, as described in this section. With IPv6, if you want to send a frame or packet to all nodes in the local network, you use a special all-nodes IPv6 multicast address. This completely eliminates the need for the broadcast traffic type in IPv6-only networks.

Traffic Types Summary

Table 4-3 provides a summary of the traffic types described in this chapter.

Key Topic

Table 4-3 Traffic Types Summary

Traffic Type	Description
Unicast	Traffic is one-to-one communication between a single sender and a single receiver, and this is the most common type of traffic in IP networks. It is used for point-to-point communication between devices, such as client-server interactions and most Internet browsing.
Multicast	A mechanism by which groups of network devices can send and receive data between the members of the group at one time (one-to-many), instead of separately sending messages to each device in the group.
Anycast	Anycast traffic is one-to-nearest communication from a sender to the nearest of multiple receivers. Multiple devices share the same anycast IP address, but packets are routed to the nearest (or best) destination based on routing metrics such as shortest path or lowest latency.
Broadcast	A broadcast is at the opposite end of the spectrum from a unicast. Broadcast traffic is one-to-all communication from a single sender to all devices within a specific network segment.

Summary

Here are the main topics covered in this chapter:

- The focus of this chapter was to ensure you are well versed in the most common protocols we work with in the modern network. Notice that these protocols are often recognized by their well-known port numbers. When you see 443, you should immediately think about HTTPS traffic.
- This chapter also covered the most important IP protocol types in use today, including TCP, UDP, ICMP, GRE, and IPsec.
- Finally, this chapter covered the traffic types found in computer networks today, including unicast, multicast, anycast, and broadcast.

Exam Preparation Tasks

Review All the Key Topics

Review the most important topics from this chapter, noted with the Key Topic icon in the outer margin of the page. Table 4-4 lists these key topics and the page number where each is found.

Key Topic

Table 4-4 Key Topics for Chapter 4

Key Topic Element	Description	Page Number
Table 4-1	Port Assignments for Commonly Used Protocols	86
Table 4-2	Internet Protocol (IP) Types Summary	89
Figure 4-1	Sample Unicast Transmission	90
Figure 4-2	Sample Broadcast Transmission	91
Figure 4-3	Sample Multicast Transmission	92
Figure 4-4	An IPv6 Anycast Example	93
Table 4-3	Traffic Types Summary	94

Complete Tables and Lists from Memory

Print a copy of Appendix B, "Memory Tables," or at least the section for this chapter and complete as many of the tables as possible from memory. Appendix C, "Memory Tables Answer Key," includes the completed tables and lists so you can check your work.

Define Key Terms

Define the following key terms from this chapter and check your answers in the Glossary:

anycast, Authentication Header (AH), broadcast, connection-oriented, connectionless, Domain Name System (DNS), Dynamic Host Configuration Protocol (DHCP), Encapsulating Security Payload (ESP), File Transfer Protocol (FTP), Generic Routing Encapsulation (GRE), Hypertext Transfer Protocol (HTTP), Hypertext Transfer Protocol Secure (HTTPS), Internet Control Message Protocol (ICMP), Internet Key Exchange (IKE), Internet Protocol (IP), Internet Protocol Security (IPsec), Lightweight Directory Access Protocol (LDAP), Lightweight Directory Access Protocol over SSL (LDAPS), multicast, Network Time Protocol (NTP), Remote Desktop Protocol (RDP), Secure File Transfer Protocol (SFTP), Secure Shell (SSH), Secure Sockets Layer (SSL), Server Message Block (SMB), Session Initiation Protocol (SIP), Simple Mail Transfer Protocol (SMTP), Simple Mail Transfer Protocol Secure (SMTPS), Simple Network Management Protocol (SNMP), Structured Query Language (SQL) Server, Syslog, Telnet, Transmission Control Protocol (TCP), Transport Layer Security (TLS), Trivial File Transfer Protocol (TFTP), unicast, User Datagram Protocol (UDP)

Additional Resources

IP, ICMP, UDP, and TCP: https://www.ajsnetworking.com/udp-and-tcp/

Ports and Protocols Quiz for Network+: https://www.ajsnetworking.com/ports_protocols_net_plus

Review Questions

The answers to these review questions are in Appendix A, "Answers to Review Questions."

1. What protocol is considered the de facto standard when it comes to secure access to remote systems for management purposes?
   a. Telnet
   b. SSH
   c. IPsec
   d. IMAP

2. You are interested in dynamically assigning the IP address information in your IPv4-based network infrastructure. What protocol can you use to accomplish this?
   a. DNS
   b. TFTP
   c. FTP
   d. DHCP

3. What global hierarchical system is used to resolve domain names to IP addresses?
   a. TFTP
   b. DHCP
   c. NTP
   d. DNS

4. What port and protocol are used by HTTPS? (Choose two.)
   a. TCP
   b. UDP
   c. 443
   d. 123
   e. 8080
   f. 80

5. What are the port and protocol used by Syslog? (Choose two.)
   a. TCP
   b. UDP
   c. 148
   d. 514
   e. 240

6. What IP protocol do ping and traceroute use in their operation?
   a. IPsec
   b. DNS
   c. ICMP
   d. DHCP

7. What port is used for SSH?
   a. 80
   b. 443
   c. 22
   d. 23

8. What protocol negotiates cryptographic keys and security parameters for IPsec?
   a. AH
   b. ESP
   c. IKE
   d. GRE

9. What type of traffic uses a destination Layer 3 address of 255.255.255.255 and is no longer used in IPv6 environments?
   a. Unicast
   b. Multicast
   c. Broadcast
   d. Anycast

10. Which of the following is a Microsoft protocol that allows a user to view and control the desktop of a remote computer?
    a. Syslog
    b. SNMP
    c. NTP
    d. RDP

11. Which is a mechanism by which groups of network devices can send and receive data between the members of the group at one time (one-to-many), instead of separately sending messages to each device in the group?
    a. Unicast
    b. Multicast
    c. Anycast
    d. Broadcast